`tests/dynamic/suites/memory_soak_suite.sh` — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Authored for P10. Honest-SKIPs (`topology_unsupported` / `feature_disabled_by_config`, §11.4.69) until the live `dynamic` stack + RSS hook land; parse-clean + authored today.

> Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview

§11.4.169 **memory soak** suite. It drives a sustained request soak against the live `dynamic` stack while sampling the control-plane / helper RSS, and asserts the working set does **not grow without bound** (a leak). It records min/max/first/last/mean RSS (§11.4.24 style) and PASSes only when the post-warmup last sample is within `MEM_GROWTH_PCT` of the post-warmup baseline, citing the sample series.

Prerequisites

- Source library `tests/dynamic/lib/analyzer_common.sh` + committed `tests/lib/evidence.sh`.
- `curl`, `awk`, POSIX `sh`, `date`.
- For a real run: `HELIX_DYNAMIC_STACK=1` + `HELIX_PROXY_URL`, plus `HELIX_MEM_RSS_CMD` — an operator-supplied command that prints the summed RSS (KB) of the process(es) under test (e.g. the acl-helper + healthd). Config injection §11.4.28 — no hardcoded process discovery in the suite.

Usage examples

```
# Today (no live stack / no RSS hook): honest SKIP, exit 0:
bash tests/dynamic/suites/memory_soak_suite.sh

# P10 live run:
```

```
HELIX_DYNAMIC_STACK=1 HELIX_PROXY_URL=http://127.0.0.1:53128 \
  MEM_REQUESTS=500 MEM_WARMUP=50 MEM_GROWTH_PCT=20 \
  HELIX_MEM_RSS_CMD='ps -o rss= -p "$(pgrep -f external_acl)"' \
  bash tests/dynamic/suites/memory_soak_suite.sh
```

Env: `MEM_REQUESTS` (default 500), `MEM_WARMUP` (default 50), `MEM_TARGET` (default `http://target-a.internal/`), `MEM_GROWTH_PCT` (max allowed last-vs-baseline growth %, default 20).

Edge cases

- **Live stack absent** → `topology_unsupported` SKIP, exit `0`.
- **RSS hook not configured** (`HELIX_MEM_RSS_CMD` empty) → `feature_disabled_by_config` SKIP, exit `0`.
- **Growth above `MEM_GROWTH_PCT`** (or no post-warmup samples) → `FAIL` citing `memory.evidence`.

§11.4.115 RED_MODE polarity

`RED_MODE`	What it asserts	PASS means
`0` (default)	GREEN guard	working set bounded (growth ≤ `MEM_GROWTH_PCT`)
`1`	RED baseline	the pre-fix stack shows **unbounded growth** (growth > threshold) → leak reproduced

A `RED_MODE=1` run where the working set stayed bounded is a finding (`no leak to reproduce`), not a pass.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean (`sh -n` + `bash -n`, §11.4.67).
- Soaks `MEM_REQUESTS` proxied requests, sampling RSS every 10 requests into `rss.series`; `awk` computes baseline (first post-warmup sample), last, min/max/mean, and growth %.
- GREEN requires `baseline > 0` AND `growth ≤ MEM_GROWTH_PCT`. Evidence (`memory.evidence` + the series) under `qa-results/p9-harness/memory_soak_<run-id>/`.
- Resources: shell+curl only; bounded request count; §12.6 60% host ceiling.

Related

- `tests/dynamic/lib/analyzer_common.sh` — sourced base.
- `tests/dynamic/suites/run_all_suites.sh` — orchestrates this suite.
- Constitution §11.4.169 / §11.4.24 / §11.4.85 / §11.4.69 / §11.4.115 / §12.6; design spec §13.

Last verified

2026-07-01 — run with no live stack: honest SKIP, exit `0`; `sh -n` + `bash -n` parse-clean. Live soak + RSS sampling is exercised in **P10**.